

ASSEMBLY AI VOICE AGENT HACKATHON / TEAM DEVELARPER

THE LANTERN

A multilingual AI maître d' that takes a full table's order by voice, remembers every word, never invents a dish, and keeps the kitchen in perfect sync.

Powered by AssemblyAI | Running on local AI

UNDERSTANDING

LISTENING

SPEAKING

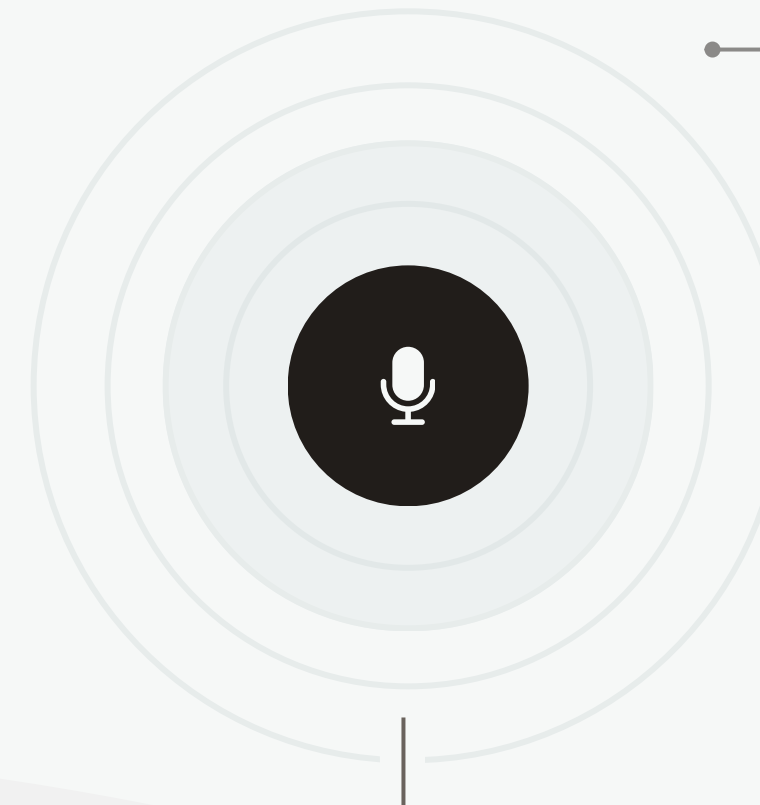


Table 12 / Order

● LIVE

01	x	Salmon	PLACED
02	x	Truffle	SYNCED
03	x	Steak	READY

Our Team



Trong Dang Thai

UI/UX Designer &
Project Presenter



Long Quan Ton

Project Lead & AI Engineer
(System Architect)



Tuan Khoa Vi

UI/UX Designer &
Project Presenter



Nomura Yoshio

AI Engineer
(Logic and Reliability)



Khanh Tuong Huynh

AI Engineer
(Evaluation and Infrastructure)

The F&B Reality

Four pressure points The Latern is designed to solve.

62%

● SAFFING GAP

of restaurant owners report difficulty hiring enough service staff

3-5%

● ORDER-ERROR LOSS

of revenue can be lost to incorrect orders

>70%

● ONE BAD EXPERIENCE

of diners may not return after a bad experience, wrong order or allergy issue

3.0s

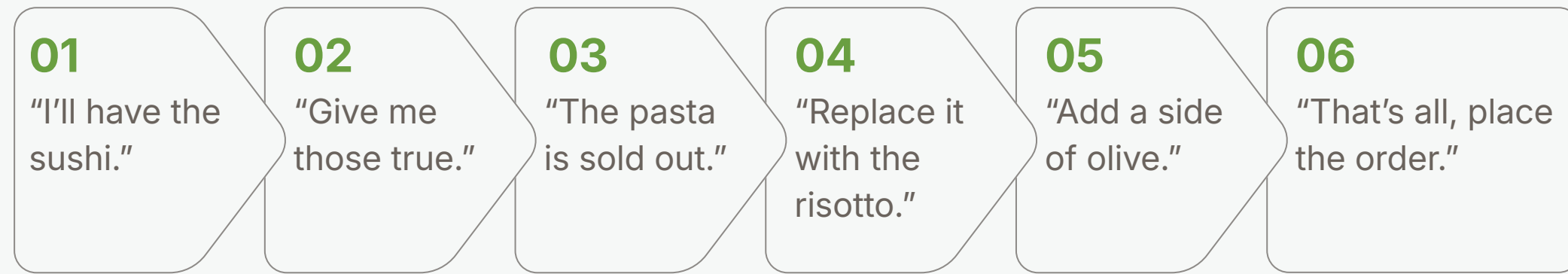
● SAFFING GAP

before users become impatient, interrupt or disengage

The Problem We're Solving

Real restaurant ordering is difficult because of multi-turn anaphoric context - customer talk in natural, fragmented ways, relying on context, references and substitutions.

A REAL ORDERING CONVERSATION



LOCAL LLM + TOOL CALLING (V1)

0 / 12

SUCCESSFUL RUNS

COMMON ERRORS

- Claimed item added but wasn't
- Accidentally deleted items
- Cancelled the order when the guest said to place it

CLOUD VOICE BOTS

19 - 86s

PER TURN LATENCY

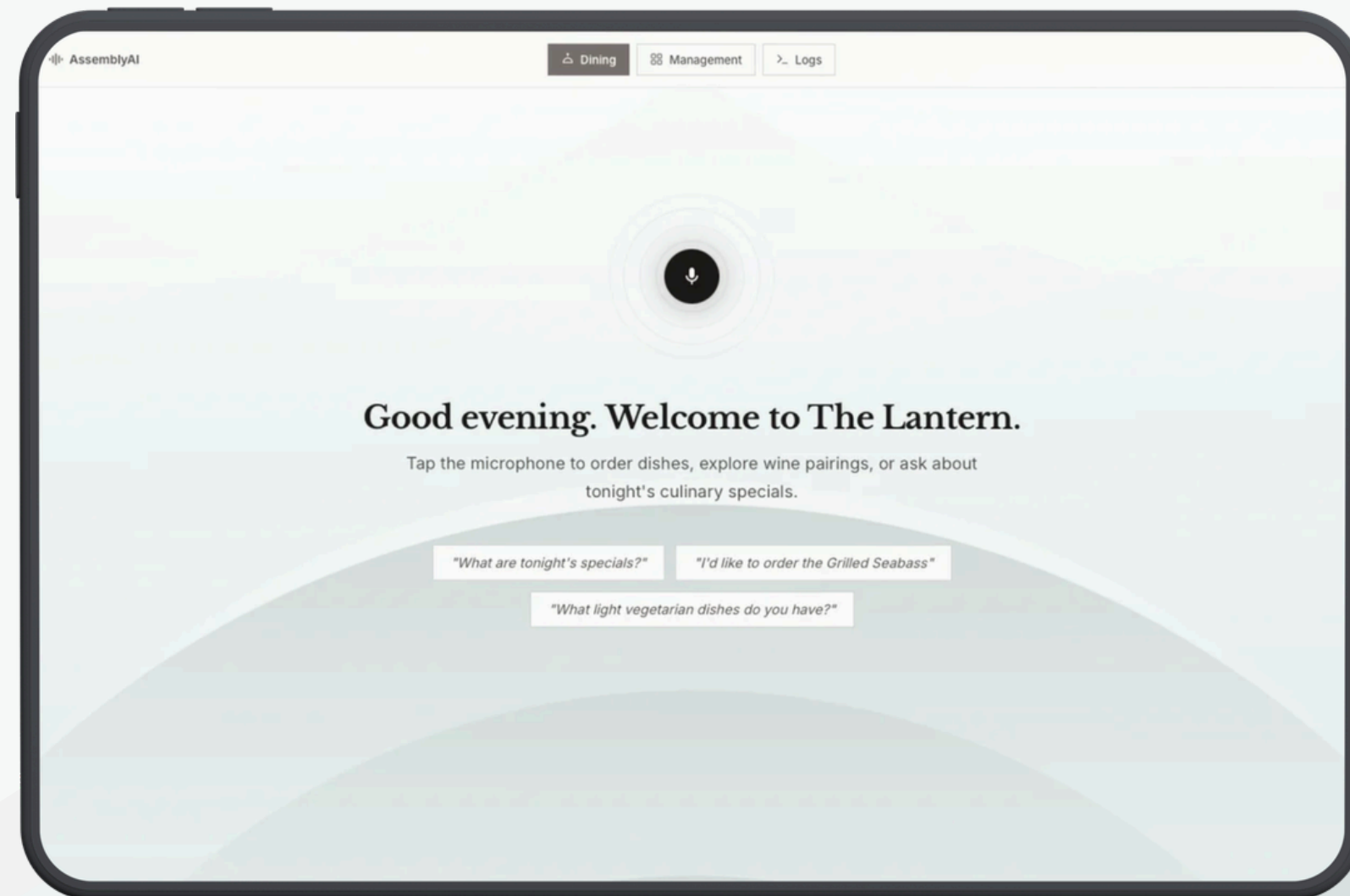
 High token & TTS costs

 Dinning-table audio privacy

The Lantern - The Multilingual AI maître d'

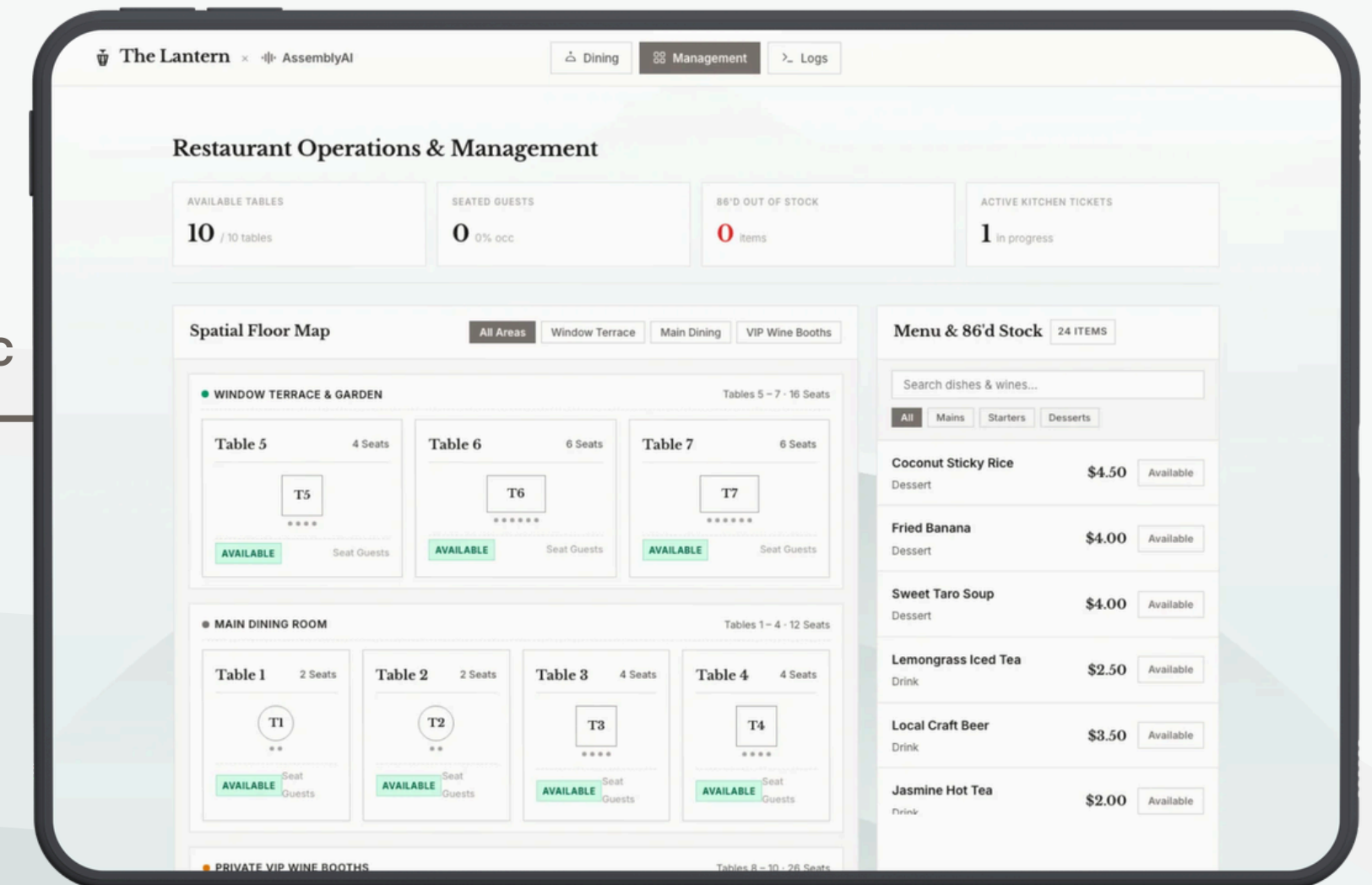
The Model Interprets; Code Decides

FOR GUESTS



ORDER SYNC

FOR RESTAURANT OPERATIONS



Natural conversation Multilingual voice, contextual references and no dead air

Controlled operations Confirmed orders, allergy, live availability and near-zero API cost

4 Pillars That Remove Order Hallucinations



Persistent Memory with SQLite

Context lives in SQLite, not in the prompt. The prompt stays fixed at ~1.46k tokens across 50 turns.



Deterministic Resolver

Resolves "those two" & "instead" against known SKUs. Rejects phantom modifiers & allergy conflicts.



Kitchen Display System

Only "place order" reaches the. 86'd items and cooking status return to the table in the next.



Zero Dead Air

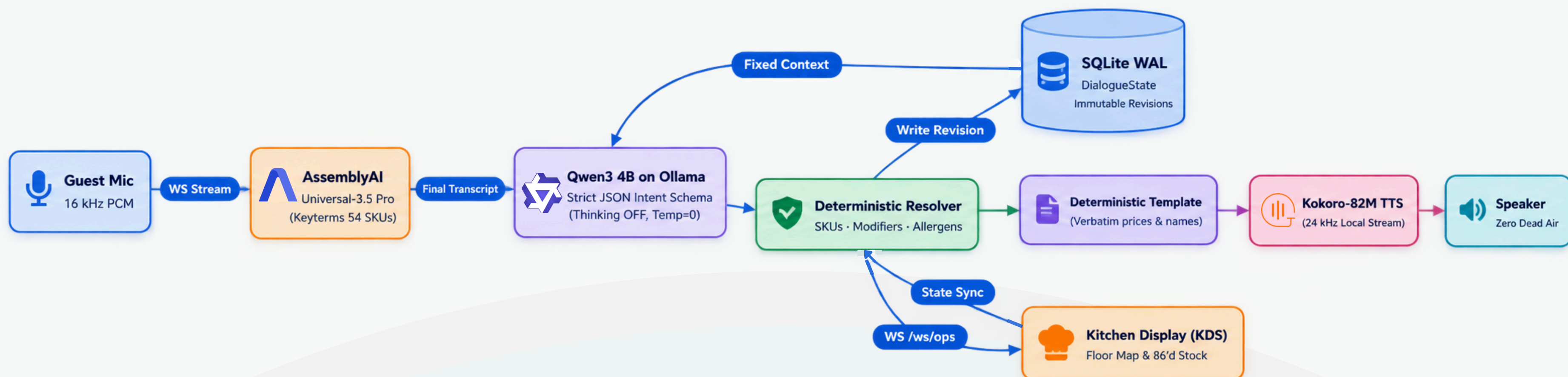
A natural filler plays in under 50 ms while Qwen and Kokoro prepare the full response.

The Lantern vs Existing Solutions

Sources (accessed 28 Sep 2026): Toast Support, Mobile Order & Pay (2026); Presto, Voice AI; AssemblyAI, Models & Streaming Billing

CRITERIA	QR / TABLE MENU Toast / Tabler	DRIVE-THRU BOT Presto / Hi Auto	GENERIC VOICE AGENT Cloud LLM	THE LANTERN V2 Local edge control
GUEST EXPERIENCE	Manual tapping harder for elderly guests	Rigid dialogue Built for simple drive-thru orders	Natural speech Can mishear items or invent prices	Natutal table conversation Multilingual with instant voice fillers
6-TURN ORDER ACCURACY	Customers enter each item	MEDIUM Static flows	0% 0/12 benchmark runs	100% 10 / 10 benchmark runs
RESPONSE LATENCY	N/A	3-6s	19 - 86s	~1.7 - 2.6s
API COST / 1000 ORDERS	\$0	\$50 - \$100	\$35 - \$60	<\$1 STT only
AUDIO / DATA PRIVACY	No voice data	Full audio sent to cloud	Full audio sent to cloud	Local LLM + order state Cloud STT only

Hybrid Architecture



Technical Highlight



Universal-3.5 Pro listens with 54 menu keyterms

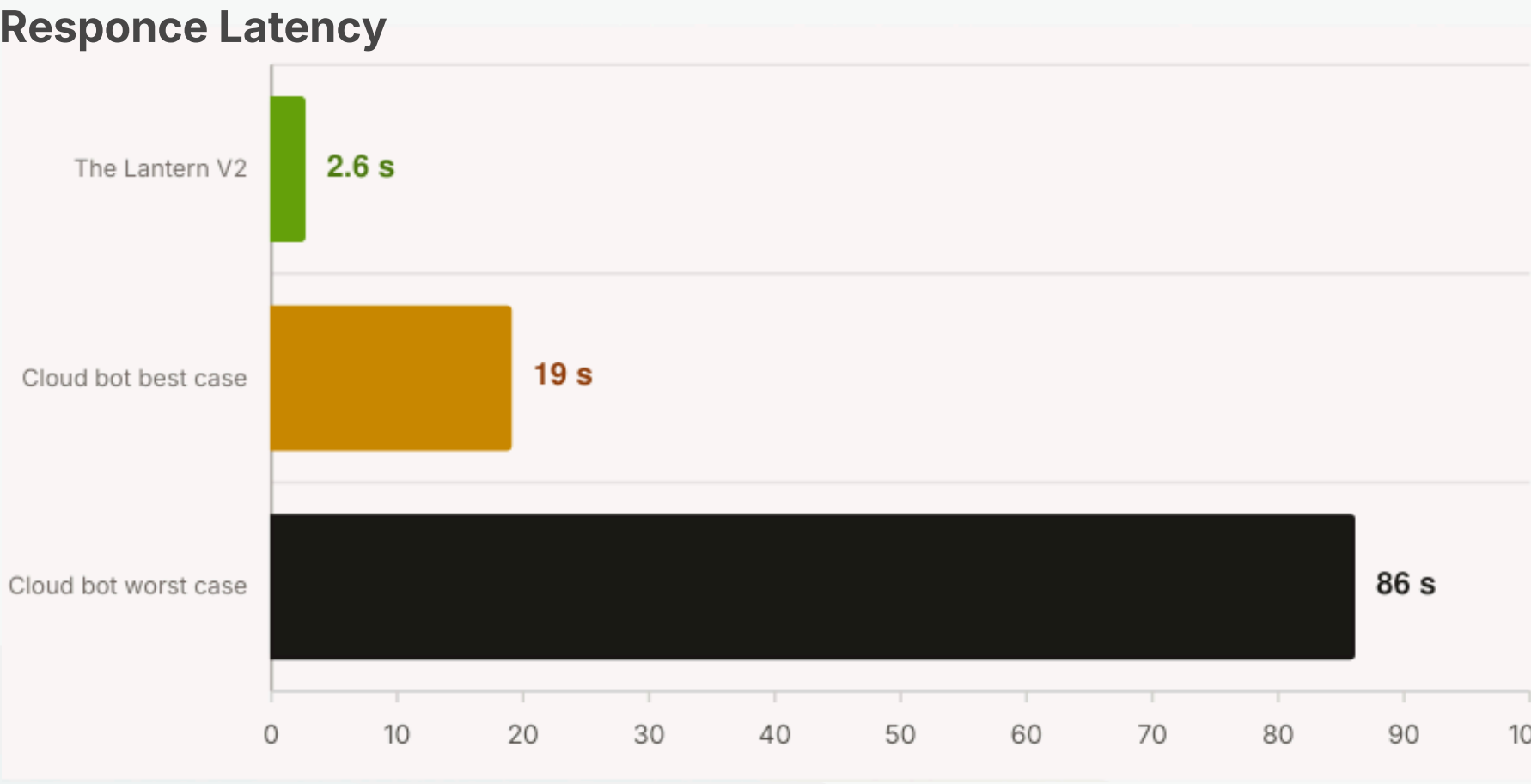
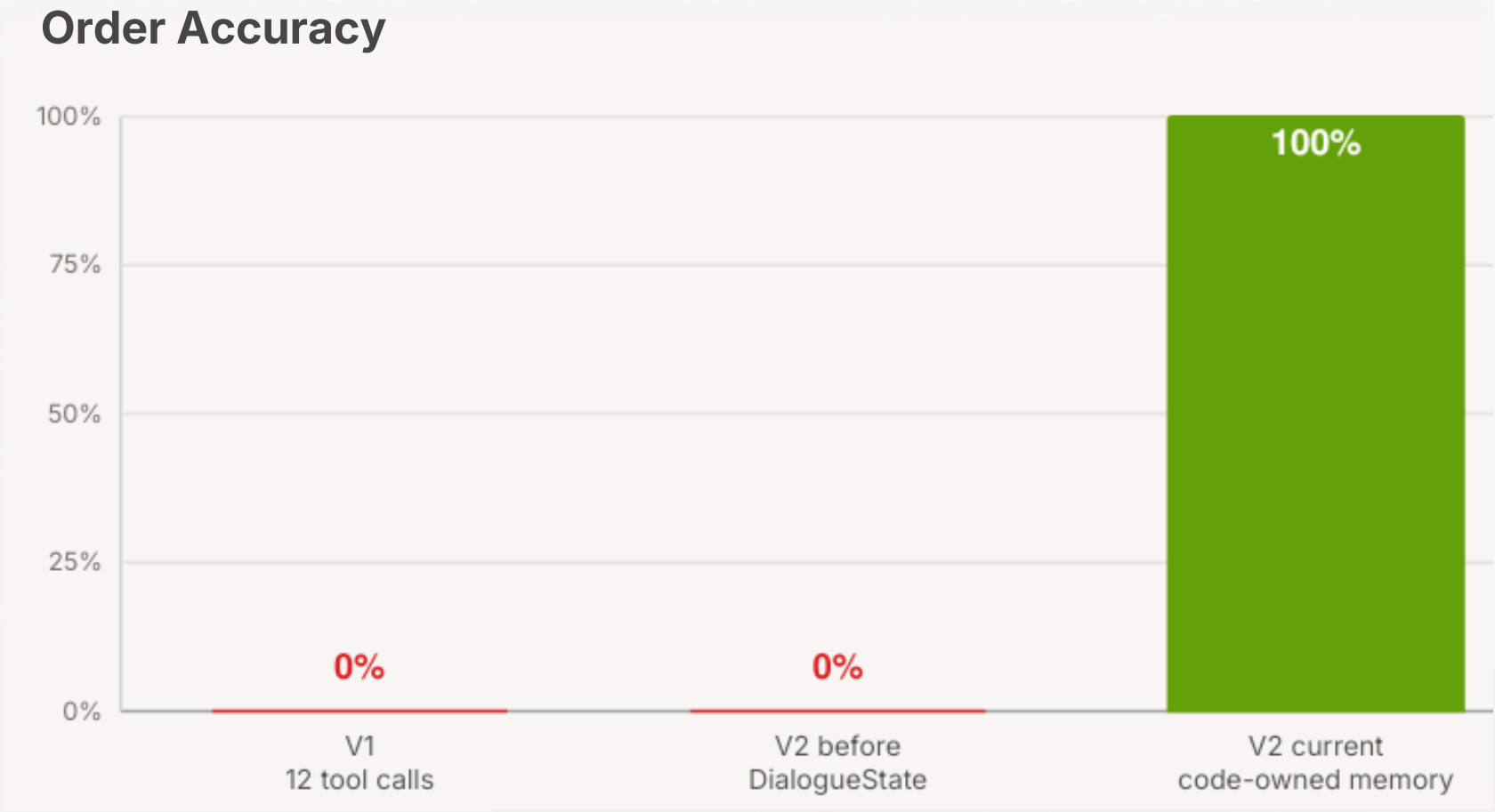


Intent and Reference only, no free-form answer

SQLite (WAL mode)

Order history survives refresh and service restarts

Verified Benchmarks



10 / 10 PRIMARY RUN ON RTX 3060
Every six-turn order matched the expected cart

1.059s QWEN INTENT
47,6% FASTER

1.752s FIRST FULL AUDIO
~50ms FILLER

REALITY CHECK

5 / 5 CROSS-HARDWARE RUN
RTX 5060 and Macbook validation

3 / 3 REFRESH RECOVERY
Orders survive an F5 mid-session

~1.46k TOKENS HELD FLAT
Turn 1 through 50 of a 4,096-token window

50 AUTOMATED CI TESTS
All safety gates covered

B2B Business Model

First Customers

- Tourism Dining**
Restaurants in US, Bangkok or Tokyo serving multilingual guests
- Restaurant Chains**
Mid to premium operators facing high waiter turnover
- Hotels & Resorts**
In-room dining that needs consistent voice operating

How Revenue Works

One-Time Setup
EDGE KIT
Edge computer, table devices, installation and onboarding

Proposed SaaS
\$29-\$49 Per Active Table / Month
Maintenance, menu updates, KDS and analytics

Owner ROI

+30% SERVICE SPEED Higher table turnover	2.5x TABLE PER WAITERS More coverage with the same floor team
100% WRONG-ORDER LOSS Comperisation leakage remove	<8 mo PAYBACK PERIOD Initial investment recovery

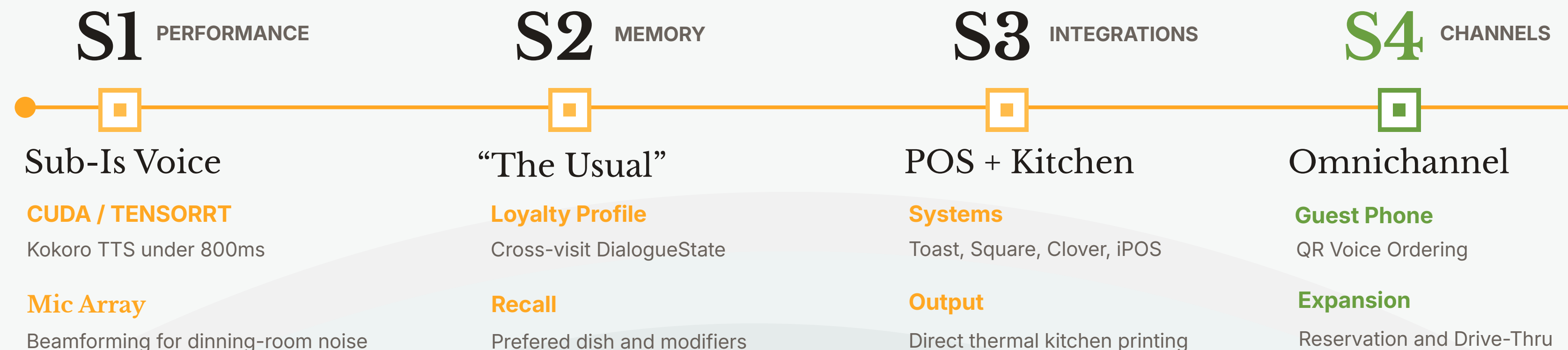
Sources (accessed 28 Sep 2026): Sinqro, Table App Pricing; CUKCUK, Pricing; AssemblyAI, Models & Streaming Billing

CALCULATION EXAMPLE

20 Tables × **\$29-\$49** = **\$580-\$980**
PER TABLE / MONTH MONTHLY SaaS REVENUE

Software subscription only
Edge hardware and setup billed once

From Table Voice Agent To AI F&B OS



TARGET STATE

AI F&B OS

One AI Operating layer for table service, kitchen operation and off-premise ordering